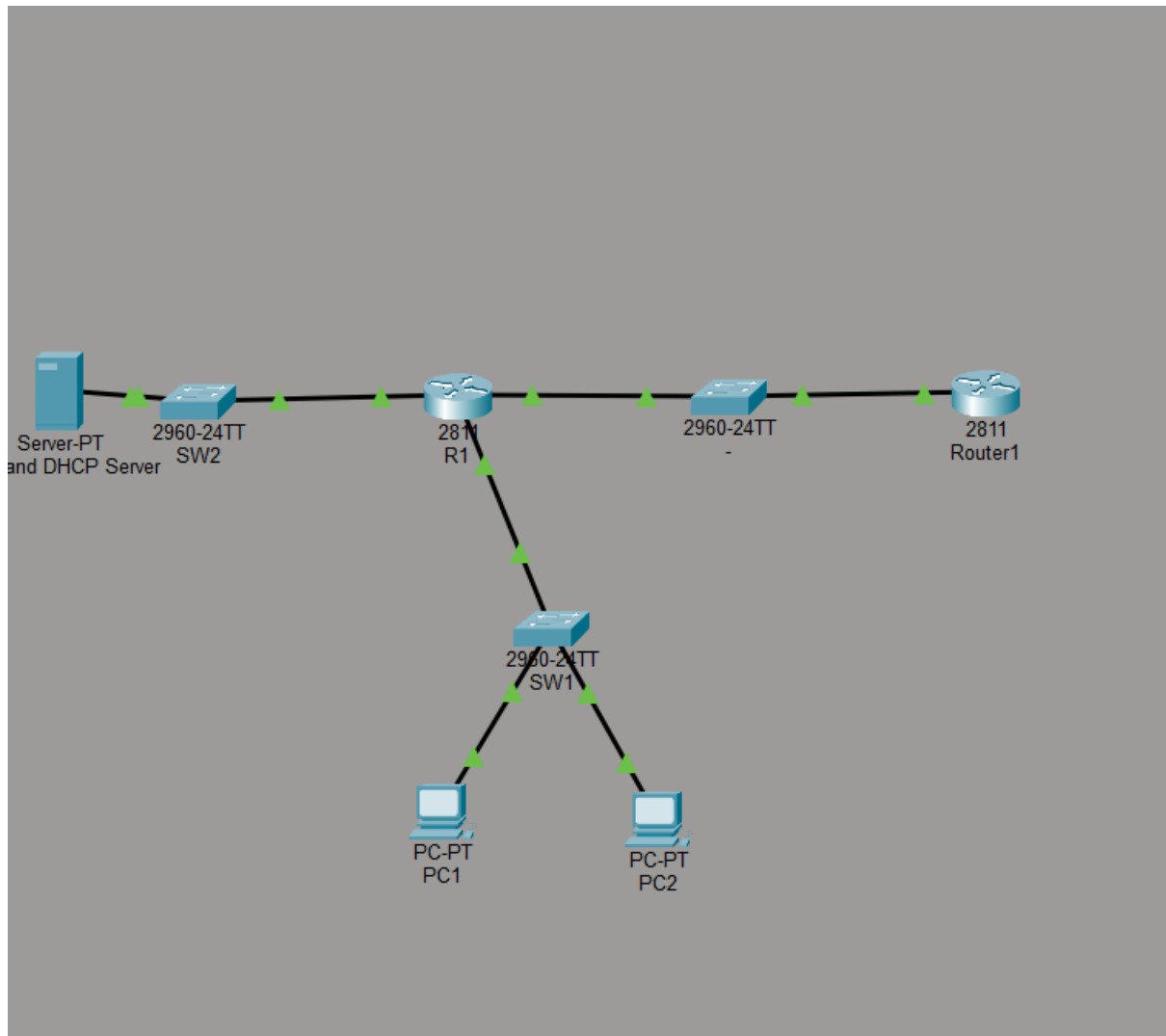


Jay Jay Hanekom Week 10 Packet tracer practical



Task 97

```

Router3
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 203.0.113.1
Router(config)#ip dhcp pool OUTSIDE
Router(dhcp-config)#network 203.0.113.0
% Incomplete command.
Router(dhcp-config)#network 203.113.0 255.255.255.0
^
% Invalid input detected at '^' marker.

Router(dhcp-config)#exit
Router(config)#ip dhcp pool OUTSIDE
Router(dhcp-config)#network 203.0.113.0 255.255.255.0
Router(dhcp-config)#default-router 203.0.113.1
Router(dhcp-config)#

```

Task 98

```

R1
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed
state to up
ip address 10.10.20.1 255.255.255.0
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
%SYS-5-CONFIG_I: Configured from console by console

R1(config)#int fa0/0
R1(config-if)#ip address dhcp
R1(config-if)#
%DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet0/0 assigned DHCP address
203.0.113.2, mask 255.255.255.0, hostname R1

R1(config-if)#ip dhcp excluded-address 10.10.10.1 10.10.10.10
R1(config)#ip dhcp pool CAMPUS_LAN
R1(dhcp-config)#network 10.10.10.0 255.255.255.0
R1(dhcp-config)#default-router 10.10.10.1
R1(dhcp-config)#dns-server 10.10.20.10
R1(dhcp-config)#

```

PC1

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 10.10.10.11

Subnet Mask: 255.255.255.0

Default Gateway: 10.10.10.1

DNS Server: 10.10.20.10

Task 99

```
R1(dhcp-config)#no ip dhcp pool CAMPUS_LAN
R1(config)#int fa0/1
R1(config-if)#ip helper-address 10.10.20.10
R1(config-if)#
```

DNS and DHCP Server

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: Pool1

Default Gateway: 10.10.10.1

DNS Server: 10.10.20.10

Start IP Address: 10 10 10 1

Subnet Mask: 255 255 255 0

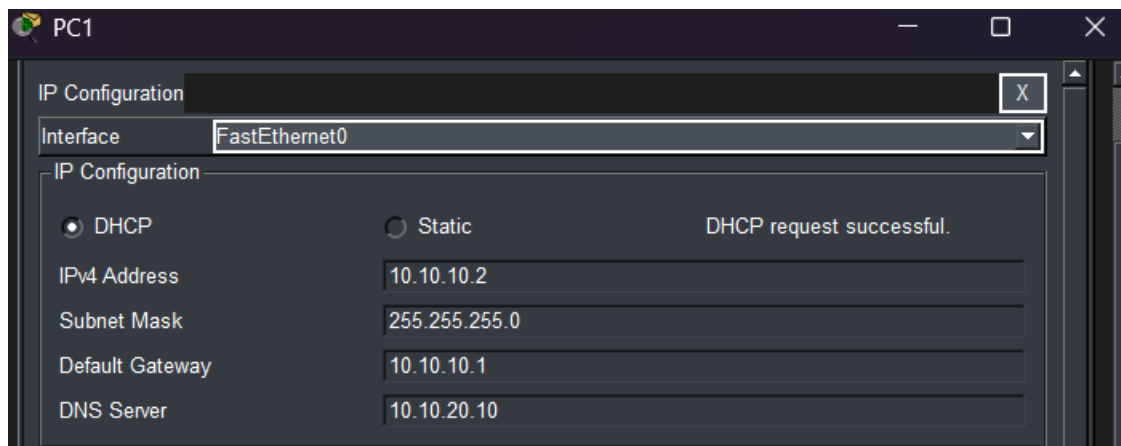
Maximum Number of Users: 100

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Pool1	10.10....	10.10....	10.10....	255.2...	100	0.0.0.0	0.0.0.0



Task 100

Client DHCP discovers start as a broadcast, which the router will not forward, because the server is on a different subnet the discover won't reach it. The Ip helper address catches the broadcast on the router, and forwards it towards the server.

If its on the same subnet then it would not have an issue because it wouldn't need to go through a different subnet.